Weekly Progress Report: GrantLens

Course Code : CSE299 | Sec : 3

Project Group No: 3

Group Members:

Anas Ibne Majbha (2322846642)
Tazreen Alam (2321506042)
Hasin Anzum Inan (2323133042)
Israt Jahan Susmi (2112280642)

Week 3: Gemini AI API Integration, Full-Stack Portal & Docker Setup

Sprint Goal: Transition GrantLens from structural parsing into an intelligent qualitative evaluation platform. Integrate Google Gemini 1.5 Flash via official GenAI SDK to simulate an authoritative NSF/NIH Scientific Review Officer (SRO) panel chair. Deploy a decoupled FastAPI REST gateway with strict Pydantic schemas and stateless JWT security, upgrade the React Researcher Portal with multi-stage progress steppers and interactive SRO scorecards, deploy production multi-container Docker orchestration (FastAPI + PostgreSQL 16), and engineer high-efficiency database response caching reducing repeat audit retrieval latency to under 15ms.

Planned Objectives

Member	Role	Weekly Objectives
Hasin Anzum Inan ID: 2323133042	External Gemini AI Layer & Context Pipeline	Google GenAI SDK integration with Gemini 1.5 Flash; SRO panel chair system prompt engineering; 4-pillar rubrics (Intellectual Merit, Broader Impacts, Alignment, Budget); context window token budgeting (14k solicitation / 28k proposal narrative); strict native JSON mode enforcement; resilient mock fallback engine; interactive on-demand paragraph rewriter.
Anas Ibne Majbha ID: 2322846642	Backend REST Gateway & Security Lead	FastAPI asynchronous non-blocking routing core under /api/v1; RFC 6749 stateless JWT authentication guards with strict tenant isolation (get_current_user); Pydantic v2 DTO contracts (AIReviewResponse, AIRewriteRequest/Response); cross-service orchestration linking PDF extraction, caching, and Gemini API calls.
Tazreen Alam ID: 2321506042	Client-Side API Consumer & Portal UI Lead	React network architecture with centralized IPv4 loopback (127.0.0.1:8000); automatic JWT Bearer header injection; silent session probe (/auth/me); 3-stage animated progress stepper overcoming 3-5s LLM latency; interactive SRO Scorecard (composite ring gauge, verdict badges); side-by-side diff explorer and live rewriter playground.
Israt Jahan Susmi ID: 2112280642	Docker DevOps & Response Caching Lead	Multi-container Docker Compose setup (FastAPI + PostgreSQL 16 Alpine); hardened Dockerfile with native C++ compilation headers (libmupdf-dev, libpq-dev) and non-root user (grantlens UID 10000); relational AIReview schema; sub-15ms response caching (99.5% latency drop); automated test suite (test_week3_ai_pipeline.py) and telemetry.

Progress Summary

Work Completed:

- Integrated **Google Gemini 1.5 Flash** via official google-genai SDK, leveraging 1M token context window and sub-second inference latency.
- Engineered **Scientific Review Officer (SRO) persona** evaluating 4 core rubrics: Intellectual Merit, Broader Impacts, Solicitation Alignment, and Budget Feasibility (0-100%).
- Implemented **Context Token Budgeting**: algorithmic slicing of 14,000 character solicitation rules and 28,000 character proposal narrative with explicit page markers.
- Enforced **Native JSON Output Schema** (response_mime_type="application/json", temp 0.2) paired with a resilient pre-computed offline fallback simulation fixture.
- Deployed **Decoupled FastAPI REST Gateway**: asynchronous non-blocking routing, RFC 6749 JWT token security, and strict tenant ownership verification.
- Authored **Pydantic v2 DTO Contracts**: AIReviewResponse, RubricScoresSchema, and AIRewrite DTOs guaranteeing zero undefined properties in client UI.
- Upgraded **React Researcher Portal**: centralized IPv4 client config, silent session probe (/auth/me), and a 3-stage animated progress stepper resolving LLM latency.
- Delivered **Interactive SRO Scorecard & Playground**: circular SVG composite ring gauge, panel verdict badges, side-by-side diff viewer, and live paragraph rewriter.
- Engineered **Relational Response Caching** in PostgreSQL: intercepting audit calls to return stored evaluations in <15ms at $0.00 cost (99.5% latency reduction).
- Containerized production environment via **Docker Compose**: PostgreSQL 16 Alpine, hardened backend Dockerfile (non-root UID 10000, C++ build headers), and health checks.
- Validated system with **Automated End-to-End Test Suite** (test_week3_ai_pipeline.py) passing all 6 phases, and built live /api/v1/status telemetry dashboard.

Challenges Encountered:

- Generative AI inference latency: Evaluating 40,000 characters takes 3-5 seconds, creating perceived freezing when using generic static spinners.
- LLM output formatting inconsistencies: Freeform conversational LLM text or markdown backticks broke automated JSON parsers.
- Redundant API billing & quota exhaustion: Repeatedly evaluating identical manuscripts caused excessive token costs and latency.
- Docker container startup race conditions: The FastAPI web container attempted database connections before PostgreSQL finished initializing.

Proposed Solutions:

- Engineered a 3-stage animated progress stepper in React (Document Parsing ➔ SRO Deliberation ➔ Scorecard Synthesis), eliminating wait fatigue.
- Configured Gemini's native response_mime_type="application/json" with low temperature (0.2) and strict Pydantic validation schemas.
- Implemented gateway-level response caching in PostgreSQL, checking existing reviews first and returning cached results in <15ms.
- Configured Docker Compose depends_on with condition: service_healthy using pg_isready healthcheck retries before starting the web backend.

Overall Status:

- All Week 3 milestone deliverables completed with 100% test coverage across all 6 test phases. System fully operational, cached, and containerized.

Detailed Weekly Log

During Week 3, the team achieved complete full-stack integration connecting React, FastAPI, and PostgreSQL with Google Gemini 1.5 Flash. All external AI connections, asynchronous REST routes, UI scorecards, and Docker containers were deployed and verified with automated test suites.

Member 1: Hasin Anzum Inan led external LLM integration and prompt engineering — connected the google-genai SDK, engineered the SRO panel chair persona and 4 federal rubrics, built the 14k/28k token context budgeting algorithm, enforced native JSON mode, and built the offline simulation fallback.

Member 2: Anas Ibne Majbha engineered backend REST API routing and security perimeter — configured FastAPI asynchronous endpoints under /api/v1, enforced RFC 6749 stateless JWT authentication and tenant isolation, authored Pydantic v2 DTO contracts, and managed cross-service orchestration.

Member 3: Tazreen Alam architected client-side API consumption and interactive UI — developed centralized IPv4 client config, silent session verification (/auth/me), 3-stage animated progress stepper, interactive SRO Reviewer Scorecard, side-by-side diff explorer, and live paragraph rewriter playground.

Member 4: Israt Jahan Susmi managed Docker DevOps and response caching — authored the hardened Dockerfile with C++ compilation headers and non-root security user, configured docker-compose.yml with PostgreSQL 16 Alpine, engineered the sub-15ms AIReview caching engine, and authored the 6-phase test harness.

Report Date: November 2026

Supervisor Signature: _____